BETHANY HIGH SCHOOL, Sarjapur, Bangalore
REVISION TEST
Std – VIII
CHEMISTRY

TOTAL MARKS: 20 TIME: 10 MINS

Section A
(Attempt all questions from this **Section**.)

Question 1

Choose the correct answers to these questions from the given options. [4]

(Do not copy the questions, write the correct answers only.)

(i) An atom has 2 electrons in its first shell and 6 electrons in its second shell. How many more electrons does it need to complete its outermost shell?

a) 1
b) 2
c) 6
d) 8

(ii) A student observes that a substance changes colour, gives off a gas and forms a new substance when two substances are mixed. This is most likely a:

a) Physical change
b) Change of state
c) Chemical change
d) Temporary change

(iii) Which statement about hydrogen is correct?

A: It has two electrons in its first shell.
B: It shows only metallic characteristics.
C: It shows characteristics similar to both alkali metals and halogens.

a) Only A
b) Only B
c) Only C
d) Both A and C

(iv) Which of the following contains a covalent bond?

a) Sodium chloride
b) Magnesium oxide
c) Carbon tetrachloride
d) Calcium chloride

Question 2

(i) Complete the following by choosing the correct answer from the bracket. [3]

a) The particle which determines the identity of an element is the _____. (proton / electron)

b) Hydrogen can gain one electron and attain the stable noble gas structure of _____. (Helium / Neon)

c) When ammonium chloride is heated, it changes into ammonia and hydrogen chloride. On cooling, they combine again to form ammonium chloride. This is a _____ change. (reversible / irreversible)

(ii) Study the table and answer the questions. [3]

Atom	Number of Protons	Number of Neutrons	Number of Electrons
M	6	6	6
N	8	8	8
P	1	0	1

a) Which atom represents carbon?

b) Which atom has the highest number of mass number?

c) Which atom is likely to lose electrons and therefore exhibit metallic characteristics?

Section B

(Attempt ***anyone*** questions from this **Section**.)

Question 3

(i) Answer the following. [2]

a) Give two signs that can indicate that a chemical reaction has taken place.

b) State one difference between a thermal dissociation and thermal decomposition reaction.

(ii) Draw a neat orbital/shell structure of an atom having the electronic configuration 2, 8, 2. [2]

Identify whether the atom is likely to gain, lose, or share electrons.

(iii) A student observes the following changes during a science experiment: [3]

A: Calcium oxide reacts with water to form a single product.

B: A compound breaks down into simpler substances on heating.

C: Zinc reacts with copper sulphate solution and replaces copper.

Answer the following:

a) What type of chemical reaction is represented by A?

b) Identify the type of reaction represented by B.

c) Which of the above reactions involves one substance breaking down into simpler substances?

(iv) Answer the following questions based on the preparation of hydrogen. [3]

a) Name the metal used in the laboratory preparation of hydrogen.

b) Why is hydrogen collected over water during its preparation?

c) State one test to identify hydrogen gas.

Question 4

(i) State the type of bonding in the following molecules: [2]

a) Magnesium oxide

b) Methane

(ii) Complete the following reactions and balance them: [2]

a) $Zn + H_2O \rightarrow$

b) $CO + H_2 + H_2O \rightarrow$

(iii) An atom X has an electronic configuration 2, 4, while atom Y has an electronic configuration 2, 8, 7. Both atoms tend to attain a stable outer shell. [3]

a) What will atom X tend to do with its valence electron?

b) State the type of bond likely to be formed between X and Y.

c) Write the formula of the compound formed by X and Y.

(iv) Write the balanced chemical equations for the following reactions: [3]

a) Sodium hydroxide and zinc sulphate.

b) Burning of phosphorus in the presence of oxygen.

c) Zinc and potassium hydroxide.

ANSWER KEY

Section A

(Attempt all questions from this **Section**.)

Question 1

Choose the correct answers to these questions from the given options. [4]

(Do not copy the questions, write the correct answers only.)

An atom has 2 electrons in its first shell and 6 electrons in its second shell. How many more electrons does it need to complete its outermost shell?

a) 1

b) 2

c) 6

d) 8

A student observes that a substance changes colour, gives off a gas and forms a new substance when two substances are mixed. This is most likely a:

a) Physical change

b) Change of state

c) Chemical change

d) Temporary change

Which statement about hydrogen is correct?

A: It has two electrons in its first shell.

B: It shows only metallic characteristics.

C: It shows characteristics similar to both alkali metals and halogens.

a) Only A

b) Only B

c) Only C

d) Both A and C

Which of the following contains a covalent bond?

a) Sodium chloride

b) Magnesium oxide

c) Carbon tetrachloride

d) Calcium chloride

Question 2

(iii) Complete the following by choosing the correct answer from the bracket. [3]

a) The particle which determines the identity of an element is the _____. (proton / **electron**)

b) Hydrogen can gain one electron and attain the stable noble gas structure of _____. (**Helium** / Neon)

c) When ammonium chloride is heated, it changes into ammonia and hydrogen chloride. On cooling, they combine again to form ammonium chloride. This is a _____ change. (**reversible** / irreversible)

(iv) Study the table and answer the questions. [3]

Atom	Number of Protons	Number of Neutrons	Number of Electrons
M	6	6	6
N	8	8	8
P	1	0	1

a) Which atom represents carbon? **M**

b) Which atom has the highest number of mass number? **N**

c) Which atom is likely to lose electrons and therefore exhibit metallic characteristics? **P**

Section B

(Attempt ***anyone*** questions from this **Section**.)

Question 3

(v) Answer the following. [2]

a) Give two signs that can indicate that a chemical reaction has taken place.

- **Change in colour**
- **Evolution of a gas**
- **Formation of a precipitate**
- **Change in temperature**
- **Emission of light or sound** *[Any two]*

b) State one difference between a thermal dissociation and thermal decomposition reaction.

(vi) Draw a neat orbital/shell structure of an atom having the electronic configuration 2, 8, 2. [2]

Identify whether the atom is likely to gain, lose, or share electrons. **Lose**

(vii) A student observes the following changes during a science experiment: [3]

A: Calcium oxide reacts with water to form a single product.

B: A compound breaks down into simpler substances on heating.

C: Zinc reacts with copper sulphate solution and replaces copper.

Answer the following:

a) What type of chemical reaction is represented by A? **Combination reaction**

b) Identify the type of reaction represented by B. **Thermal decomposition reaction**

c) Which of the above reactions involves one substance breaking down into simpler substances?

Reaction B, because one compound breaks down into two or more simpler substances.

(viii) Answer the following questions based on the preparation of hydrogen. [3]

a) Name the metal used in the laboratory preparation of hydrogen. **Zinc**

b) Why is hydrogen collected over water during its preparation? **almost insoluble in water.**

c) State one test to identify hydrogen gas. Bring a lighted splint near the gas. **Hydrogen burns with a pale blue colour flame.**

Question 4

(v) State the type of bonding in the following molecules: [2]

a) Magnesium oxide → **Ionic bond**

b) Methane → **Covalent bond**

(vi) Complete the following reactions and balance them: [2]

a) $Zn + H_2O \rightarrow$ **$ZnO + H_2$**

b) $CO + H_2 + H_2O \rightarrow$ **$CO_2 + 2H_2$**

(vii) An atom X has an electronic configuration 2, 4, while atom Y has an electronic configuration 2, 8, 7. Both atoms tend to attain a stable outer shell. [3]

a) What will atom X tend to do with its valence electron? **four valence electrons and is likely to share electrons to attain stability.**

b) State the type of bond likely to be formed between X and Y. **covalent bond.**

c) Write the formula of the compound formed by X and Y. **XY_4**

(viii) Write the balanced chemical equations for the following reactions: [3]

a) Sodium hydroxide and zinc sulphate. **$ZnSO_4 + 2NaOH \rightarrow Zn(OH)_2\downarrow + Na_2SO_4$**

b) Burning of phosphorus in the presence of oxygen. **$4P + 5O_2 \rightarrow 2P_2O_5$**

c) Zinc and potassium hydroxide. **$Zn + 2KOH \rightarrow K_2ZnO_2 + H_2\uparrow$**